APMTH 50 Modeling Activity

In this modeling activity you and your group will develop the simplified SIR differential equations from Kermack and McKendrick's paper.

Let $S(t)$ be the number of individuals in a population **s**usceptible to the flu at time t, $I(t)$ be the number of **i**nfected individuals at time t, and $R(t)$ be the number of **r**ecovered individuals. Let N be the total number of individuals in the population. Assume that the individuals in the recovered population are immune to the flu so that a recovered individual cannot become reinfected.

1. Assume that the total population N is constant. What assumption(s) must you make in order to reasonably justify a constant total population?

2. How do you think $S(t), I(t)$, and $R(t)$ should vary with time? Sketch a graph of how you think each of these populations should vary with time.

Let's now develop equations for the rates of change of each of these populations.

3. *Rate of change of susceptibles* $S(t)$
   No one is added to the susceptible group and the only way an individual leaves the susceptible group is by becoming infected with the flu. Assume that dS/dt depends on the number of individuals susceptible, the number of individuals already infected, and the amount of contact between these two populations. Suppose each infected individual has a fixed number α of contacts per day that are sufficient to spread the disease. Then we have

$$\frac{dS}{dt} = -\frac{\alpha}{N} S(t) I(t). \tag{1}$$

   (a) Why does this equation includes a factor of $1/N$?

   (b) Where does the negative sign come from?

   (c) Introduce a new parameter β so that the above differential equation becomes the equation from the paper.

4. *Rate of change of recovereds* $R(t)$
   A fixed fraction γ of the infected population will recover during any given day. Then we have

$$\frac{dR}{dt} = \gamma I(t). \tag{2}$$

   Explain why this equation does not include a factor of $1/N$.

5. *Rate of change of infecteds $I(t)$*
   We also have

$$\frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt} = 0$$

   (a) Where does this equation come from? That is, what assumption about the model does this equation reflect?

   (b) Solve this equation for dI/dt and replace dS/dt and dR/dt by the right hand sides of (1) and (2) above.

The three differential equations from parts 3c, 4, and 5 are the model equations in Kermack and McKendrick's paper!